DH Part 2

EGR 445/545

GVSU

Plan for today

- review
- more DH
  - figure 3.15 and eqn. 3.6
- project time

DH Review

- what did we learn last time?
  - why are we going to use the DH convention?

→ almost always ignore Y axis

- do not use your middle figure

DH: get from frame $i-1$ to frame i

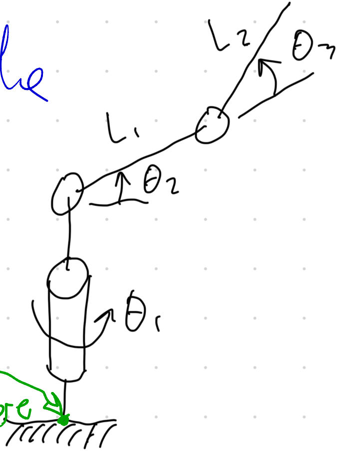
using α_{i-1}, a_{i-1}, θ_i and d_i

$\hat{X}_{i-1}$ $\hat{Z}_i$

Where does O_1 go?

… and why can't it go at the base?

O_1 goes at the intersection of z_1 and z_2



O_1 cannot go here

Order of Multiplication and Eqn. 3.6

$$ {}^{i-1}_{i}\mathbf{T} = HT_x(\alpha_{i-1}, x = a_{i-1}) \cdot HTz(\theta_i, z = d_i) $$

- matrix multiplication is not commutative
  - therefore α_{i-1} and $\cancel{ai-1}$ must come before θ_i and d_i

a_{i-1}

DH Part 2

Two Main Phases to the DH Convention

1. assign the $\hat{Z}$ axes
2. assign the origins and the $\hat{X}$ axes

- the relationship between $\hat{Z}_i$ and $\hat{Z}_{i-1}$ determines where O_{i-1} goes and where $\hat{X}_{i-1}$ points

Three Cases for Z Axes

1. Z_{i-1} and Z_i are parallel
2. Z_{i-1} and Z_i intersect
3. Z_{i-1} and Z_i are not parallel and do not intersect

Case 1: Parallel Z Axes

- this is the only case where you have some freedom in placing an origin
  - but your freedom is limited to sliding the origin along Z_{i-1}
- X_{i-1} still goes along the common perpendicular line between Z_i and Z_{i-1}
  - but now there are infinitely many common perpendiculars

Case 2: Intersecting Z Axes

- if two consecutive Z axes intersect:
  - the origin for frame $i-1$ must go at the intersection
  - X_{i-1} must be perpendicular to the plane formed by Z_{i-1} and Z_i
- **why?**

- α_{i-1} comes first

- after I rotate by α_{i-1}, my new Z axis must be // to $\hat{Z}_i$

Case 3: Non-parallel and Non-intersecting

- there is only one common perpendicular line
- X_{i-1} must go along the common perpendicular
- the origin for frame $i-1$ must go at the intersection of the common perpendicular and Z_{i-1}
  - i.e. X_{i-1} and Z_{i-1} intersect at the origin of frame $i-1$
- this case is shown in Fig. 3.15

i vs. $i-1$

- we start at frame $\{i-1\}$
- α_{i-1} and a_{i-1} refer to X_{i-1}
- θ_i and d_i refer to Z_i
- the parameter subscripts are chosen to agree with the axis subscripts
    - this will eventually feel less weird

DH Guiding Principle/Verification Question

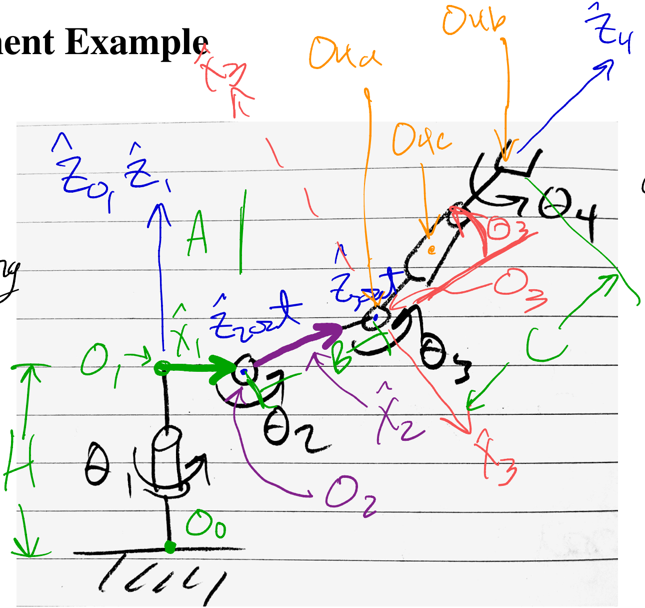
It **must** be possible to go from frame $i-1$ to frame i by following these steps **in the given order**:

1. rotate by α_{i-1}
2. translate by a_{i-1}
3. rotate by θ_i
4. translate by d_i

Frame Attachment Example

z_1 and z_2 are non-parallel, non-intersecting - case 3

$z_2 \parallel z_3$
∞ common $\perp$'s



We have ∞ choices for O_4

DH Table for previous slide

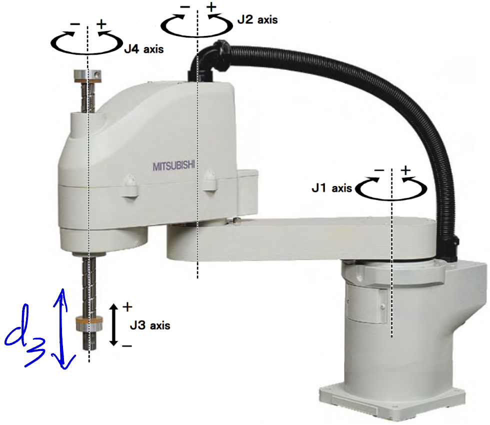
i	α_{i-1}	a_{i-1}	θ_i	d_i
1	0	0	θ_1	H
2	$+90$	A	θ_2	0
3	0	B	$\theta_3 \overset{-}{+} 90$	0
4	$\overset{-}{+}90$	0	θ_4	0 OR C

$\hat{X}_3 \perp \hat{Z}_3$ and $\hat{Z}_4$

$${}^4P_{TIP} = \begin{bmatrix} 0 \\ 0 \\ C \\ 1 \end{bmatrix}$$

$${}^4P_{TIP} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

SCARA Example



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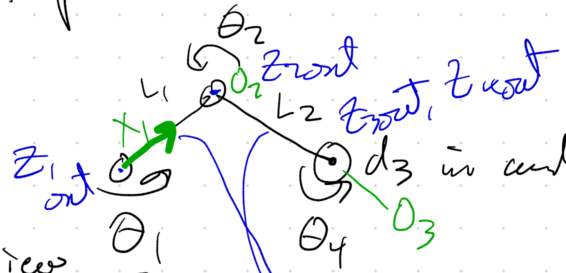
Scara

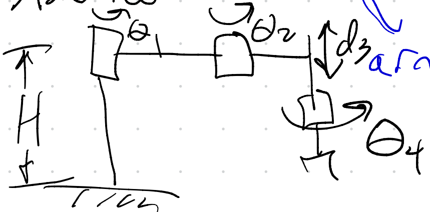
i	α	a	θ	d
1	0	0	θ_1	H
2	0	L_1	θ_2	0
3	0	L_2	0	$-d_3$
4	0	0	θ_4	0

Top View



d_3 in and out (up and down)

Side View



arms rotate in X_1/Y_1 plane

Modified SCARA 1: RRP Planar

Modified SCARA 2: 3R with Rotated Third Joint

Big Ideas for Chapter 3

- any robot can be modeled with a series of HT matrices:

$$^{0}P_{tip} = {}^{0}_{1}\mathbf{T} \cdot {}^{1}_{2}\mathbf{T} \cdot {}^{2}_{3}\mathbf{T} \cdot {}^{3}_{4}\mathbf{T} \cdot {}^{4}_{5}\mathbf{T} \cdot {}^{5}_{6}\mathbf{T} \cdot {}^{6}P_{tip}$$

- any robot link can be modeled using an HT matrix according to eqn. 3.6
  - two rotations and two translations
  - first, rotation and translation involving X_{i-1}
  - then rotation and translation involving Z_i

Key Points for Chapter 3

- Figure 3.5/3.15
- Derivation of eqn. 3.6
- DH parameter definitions
- DH Frame attachment procedure
    - the relationship between $\hat{Z}_i$ and $\hat{Z}_{i-1}$ determines where O_{i-1} goes and where $\hat{X}_{i-1}$ points

Figure 3.15

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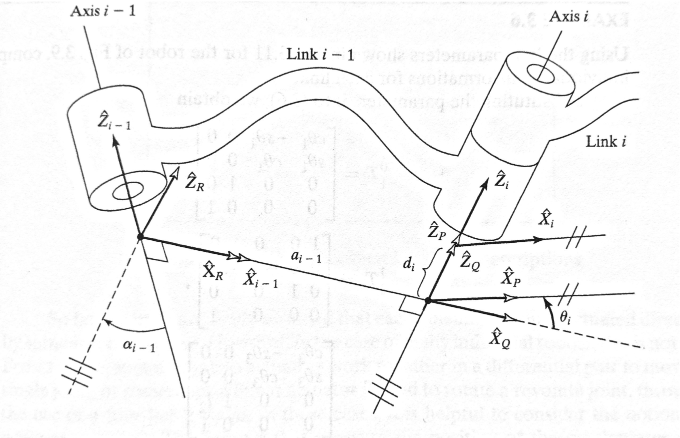
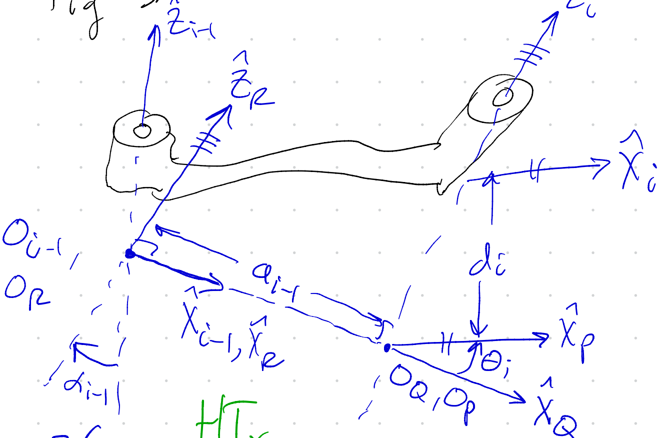


FIGURE 3.15: Location of intermediate frames $\{P\}$, $\{Q\}$, and $\{R\}$.

Fig 3.15



eqn. 3.6

$$ {}^{i-1}_{i}T = \overbrace{{}^{i-1}_{R}T \cdot {}^{R}_{Q}T}^{HT_x} \cdot \underbrace{{}^{Q}_{P}T \cdot {}^{P}_{i}T}_{HT_z} $$

- start @ frame i-1
- rotate by α_{i-1} to get to frame R
- translate by a_{i-1} to get to frame Q
- rotate by Θ_i to get to frame P
- translate by d_i to get to frame i

Figure 3.15 and Eqn. 3.6

$$ {}^{i-1}_{i}\mathbf{T} = {}^{i-1}_{R}\mathbf{T} \cdot {}^{R}_{Q}\mathbf{T} \cdot {}^{Q}_{P}\mathbf{T} \cdot {}^{P}_{i}\mathbf{T} $$

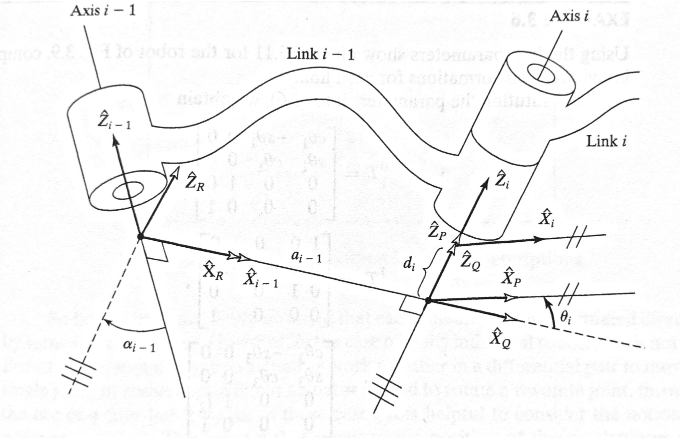


FIGURE 3.15: Location of intermediate frames $\{P\}$, $\{Q\}$, and $\{R\}$.

Eqn. 3.6

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- `part1 = robotics.HTx(alpha, x=a)`
- `part2 = robotics.HTz(theta, z=d)`
- `T = np.dot(part1, part2)`

Part 1 of Eqn. 3.6

$$\texttt{part1} = \begin{bmatrix} 1 & 0 & 0 & a_{i-1} \\ 0 & \cos(\alpha_{i-1}) & -\sin(\alpha_{i-1}) & 0 \\ 0 & \sin(\alpha_{i-1}) & \cos(\alpha_{i-1}) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- rotation by angle α_{i-1} about X
- translation by a_{i-1} along X

Part 2 of Eqn. 3.6

$$\texttt{part2} = \begin{bmatrix} \cos(\theta_i) & -\sin(\theta_i) & 0 & 0 \\ \sin(\theta_i) & \cos(\theta_i) & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- rotation by angle θ_i about Z
- translation by d_i along Z

np.dot(part1, part2)

$$
{}_{i}^{i-1}T = \begin{bmatrix} \cos(\theta_i) & -\sin(\theta_i) & 0 & a_{i-1} \\ \sin(\theta_i)\cos(\alpha_{i-1}) & \cos(\alpha_{i-1})\cos(\theta_i) & -\sin(\alpha_{i-1}) & -d_i\sin(\alpha_{i-1}) \\ \sin(\alpha_{i-1})\sin(\theta_i) & \sin(\alpha_{i-1})\cos(\theta_i) & \cos(\alpha_{i-1}) & d_i\cos(\alpha_{i-1}) \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

DH Frame Attachment Procedure

Attachment Procedure - Book

Summary of link-frame attachment procedure

The following is a summary of the procedure to follow when faced with a new mechanism, in order to properly attach the link frames:

1. Identify the joint axes and imagine (or draw) infinite lines along them. For steps 2 through 5 below, consider two of these neighboring lines (at axes i and $i+1$).

phase 2 = steps 2+4

z axes

2. Identify the common perpendicular between them, or point of intersection. At the point of intersection, or at the point where the common perpendicular meets the ith axis, assign the link-frame origin.
3. Assign the $\hat{Z}_i$ axis pointing along the ith joint axis.
4. Assign the $\hat{X}_i$ axis pointing along the common perpendicular, or, if the axes intersect, assign $\hat{X}_i$ to be normal to the plane containing the two axes.
5. Assign the $\hat{Y}_i$ axis to complete a right-hand coordinate system.
6. Assign {0} to match {1} when the first joint variable is zero. For $\{N\}$, choose an origin location and $\hat{X}_N$ direction freely, but generally so as to cause as many linkage parameters as possible to become zero.

Where does O_{i-1} go?

case 2: at the intersection of $\hat{z}_{i-1}$ and $\hat{z}_i$

case 1 & 3: at the intersection of common $\perp$ and $\hat{z}_{i-1}$

Frame Attachment Gotchas

- steps 1 and 3 are done together and are straight forward
- steps 2 and 4 trip up students
- step 2: find the origins
- step 4: assign the X axes

Origin Choices

- most common mistake: thinking you have freedom to place the origin where you want
- you only really have freedom if two consecutive Z axes are parallel

Frame Attachment Summary

- steps 1 & 3: identify the joint axes and place Z_i along them
- steps 2 & 4: identify the common perpendiculars for each pair of Z axes, Z_{i-1} and Z_i
  - depends on the Z axis case (next slide)
  - X_{i-1} along the common perpendicular for Z_{i-1} and Z_i
- step 5 (skip): Y_i to complete RH coordinate system
- step 6: Decide on frames {0} and {N}